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3rd NELIREF Data Science & AI Summer School

Introduction to Natural Language Processing (NLP), Large Language Models (LLM), and Generative AI

Instructor: Joseph Luper Tsenum

June 19-27, 2026

Outline

1. Definition of AI/ML
2. Language models
3. Definition of NLP
4. Generative AI and Attention Mechanism
5. Text presentation and representation in NLP
6. Foundation models and LLM
7. Tokenization
8. Q/A

What is Artificial Intelligence?

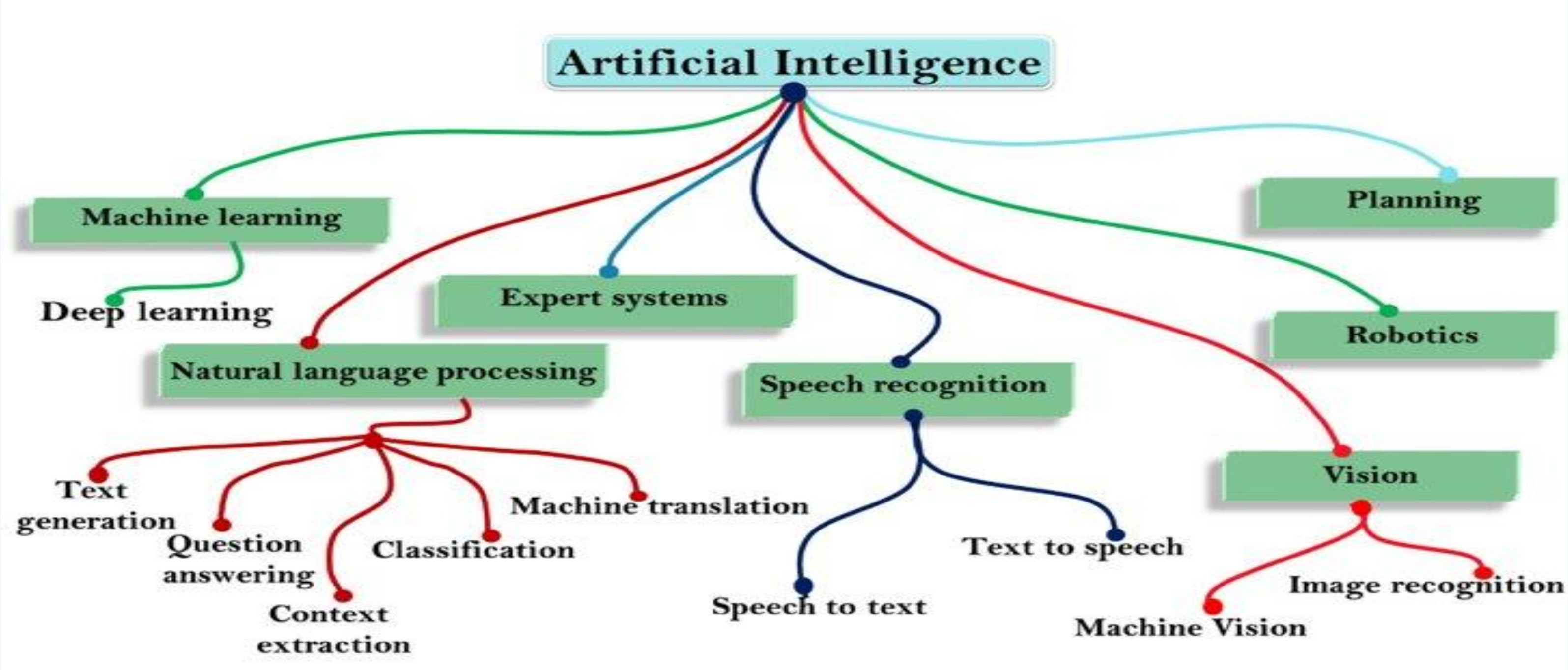
“AI is a machine’s ability to perform the cognitive functions we associate with human minds, such as perceiving, reasoning, learning, interacting with an environment, problem solving, and even exercising creativity.”

— McKinsey

You interact with AI daily without realizing it such as Siri voice assistants, customer service chatbots, Generative AI such as ChatGPT, gaming, predictive models for risk/protective factors in healthcare, forecasting models in businesses, mobile banking, fraud detection, customer engagement, social media such as Facebook, etc.

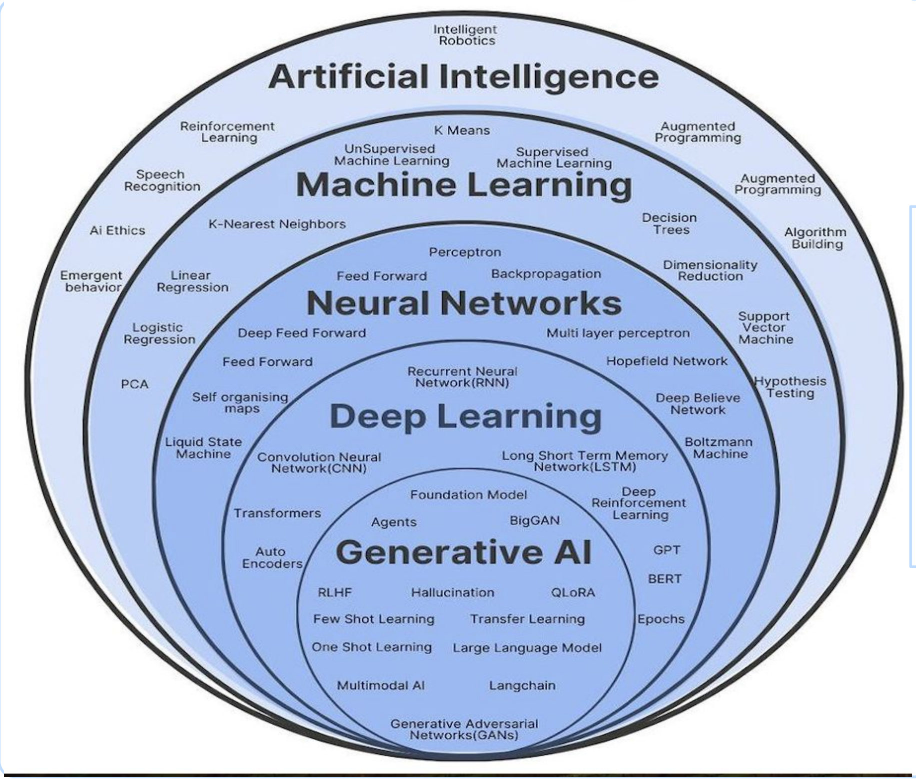
Driverless cars, scheduling of elevators, self-navigating vacuum cleaners are some of the applications of Reinforcement Learning.

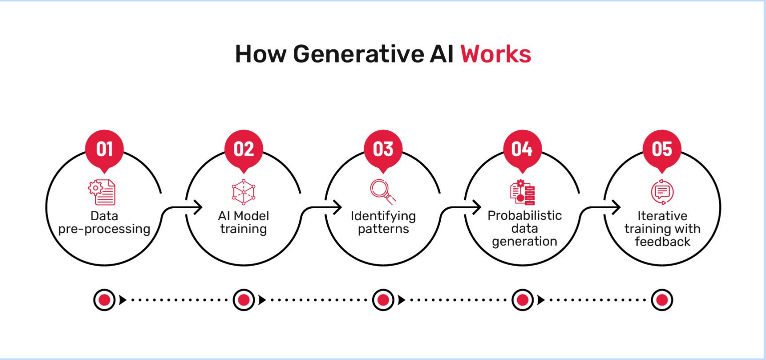
Branches of Artificial Intelligence



Source: tekrajawasthi15.medium.com — The Complete Roadmap to be a Data Scientist

AI Revolution

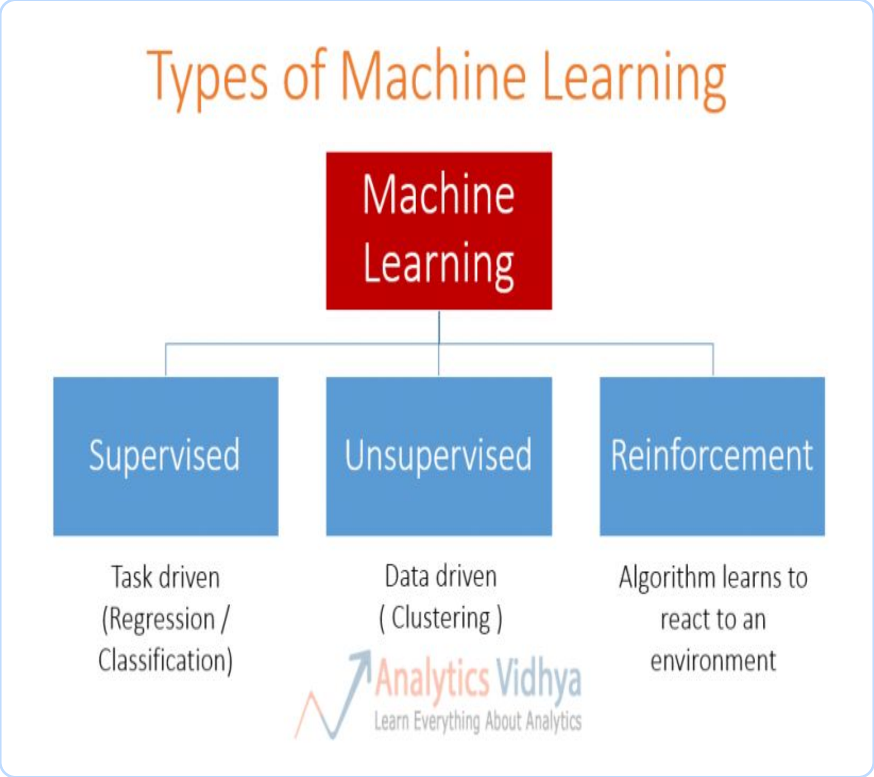




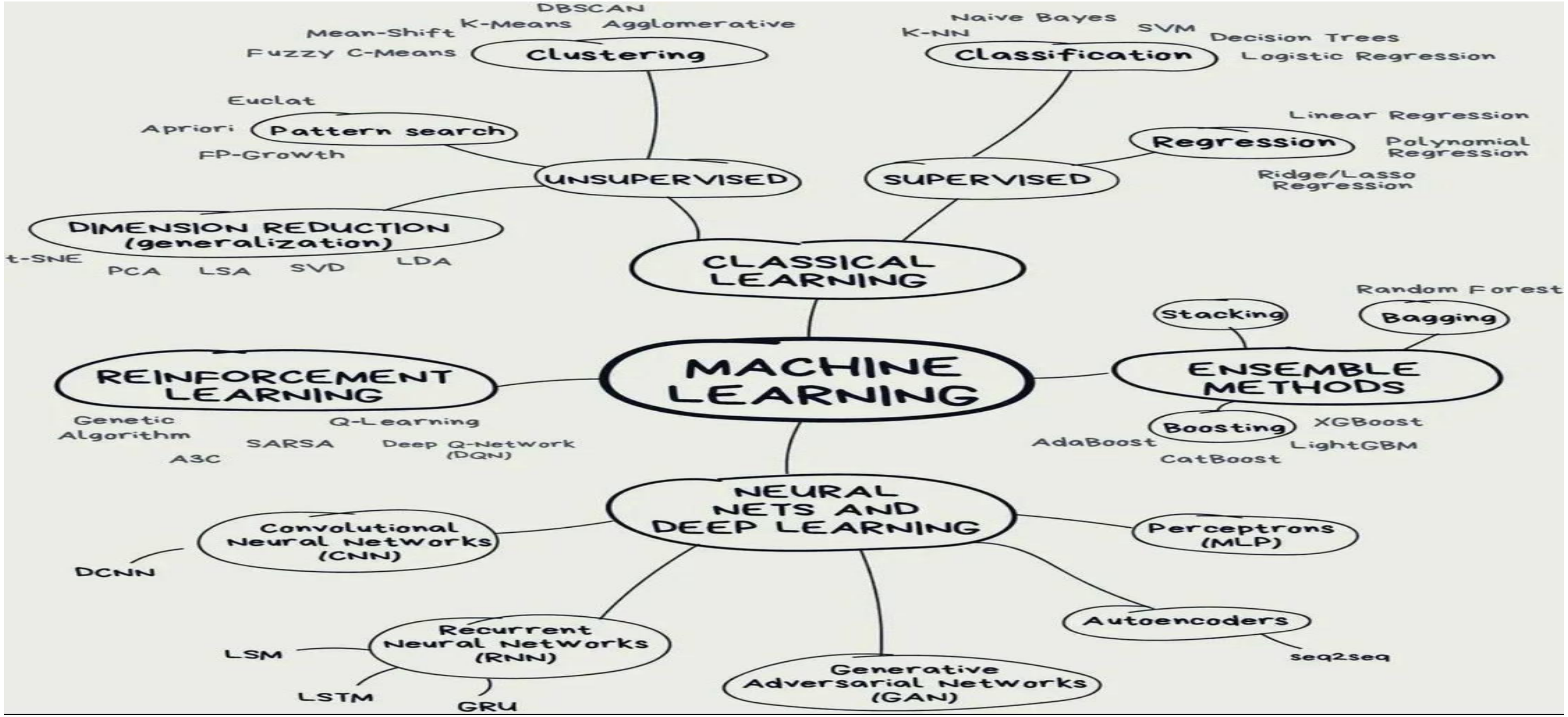
Source: https://www.analyticsvidhya.com/blog/category/generative-ai/

What is Machine Learning?

- Machine Learning is the science of teaching machines how to learn by themselves. It is a type of artificial intelligence based on algorithms that are trained on data.
- In the 1970s, machine learning was applied in medical-imaging analysis and high-resolution weather forecasting.
- Today, ML can learn from data and detect patterns thereby making predictions to help in decision and policy making.
- Deep Learning is a type of machine learning that can process large amounts of data such as images, bulk biological data, etc.
- DL requires less human intervention and can produce more accurate results (unsupervised learning) than traditional or classical machine learning methods (supervised learning). New patterns are learned.



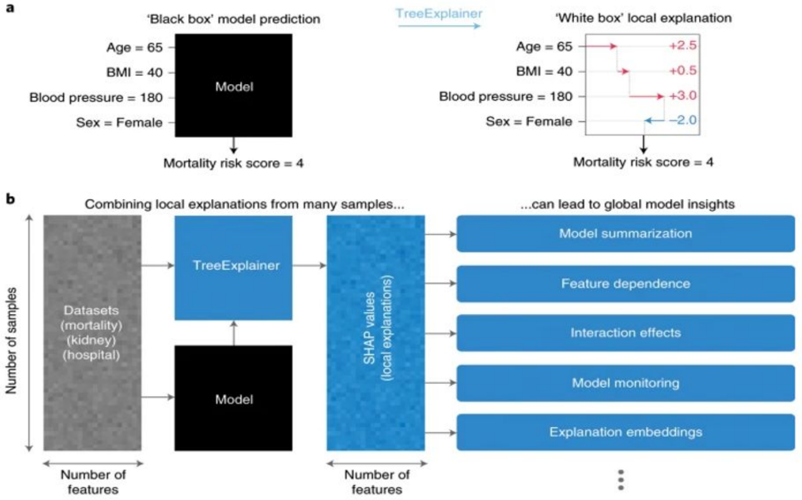
Machine Learning Methods Overview



Source: Atul Kumar

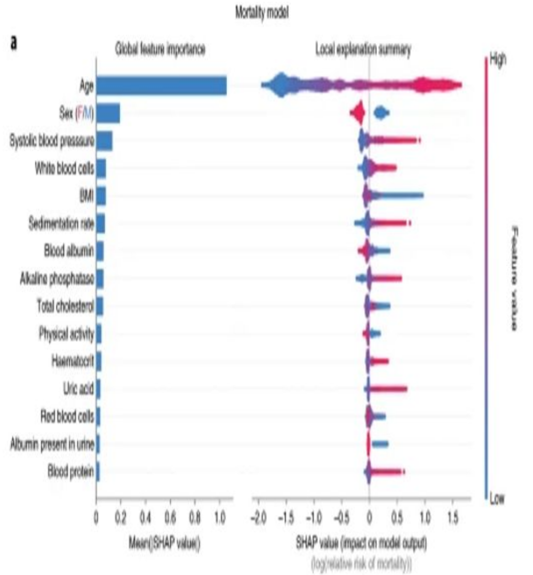
Increasing Demand for Explainable Black-Box Models

Fig. 1: Local explanations based on TreeExplainer enable a wide variety of new ways to understand global model structure.



a, A local explanation based on assigning a numeric measure of credit to each input feature. **b**, By combining many local explanations, we can represent global structure while retaining local faithfulness to the original model. We demonstrate this by using three medical datasets to train gradient boosted decision trees and then compute local explanations based on SHAP values[3]. Computing local explanations across all samples in a dataset enables development of many tools for understanding global model structure.

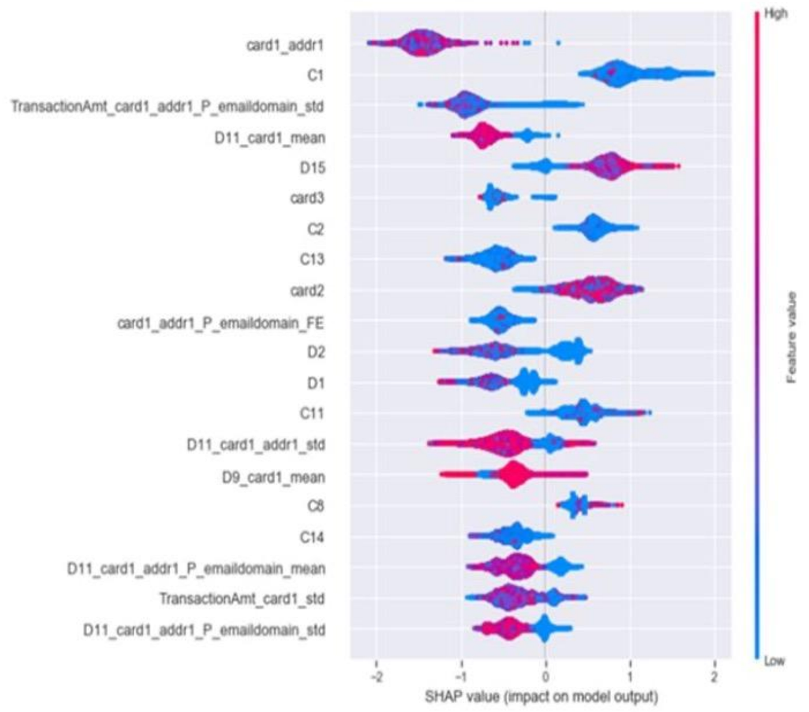
Fig. 4: By combining many local explanations, we can provide rich summaries of both an entire model and individual features.

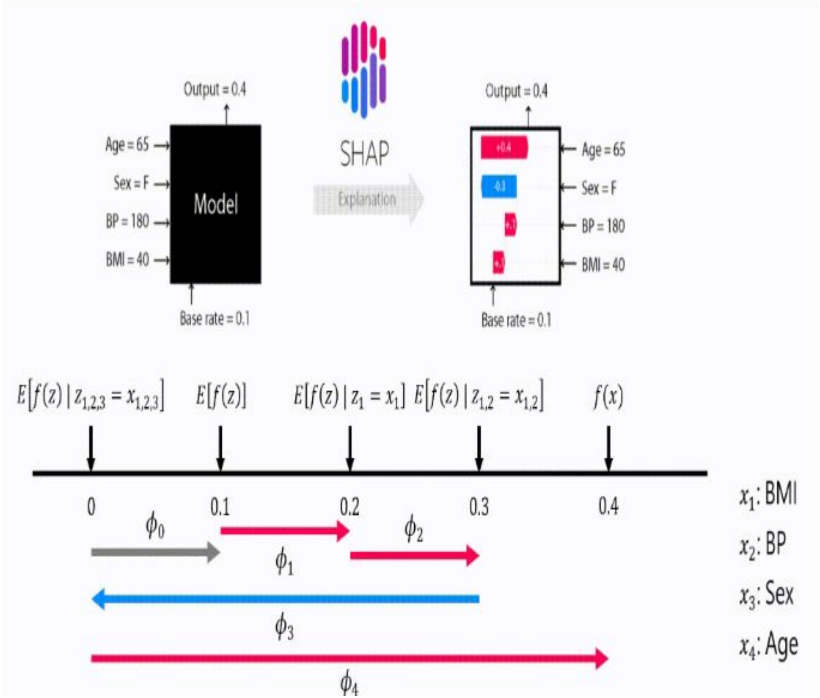


Lundberg et al., 2020

Explainable Ensemble & Deep Learning

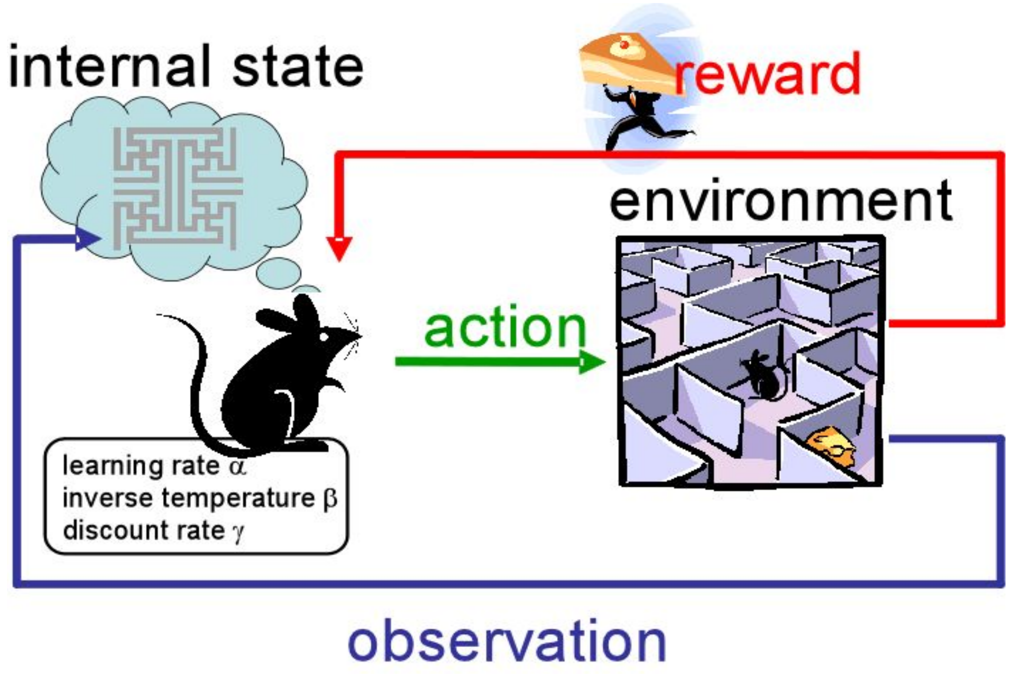
Using SHAP, LIME, and Anchors





Courtesy: Shilpi Bhattacharyya

Reinforcement Learning



Source: UTCS RL Reading Group

The pet

is the artificial agent

The good behavior

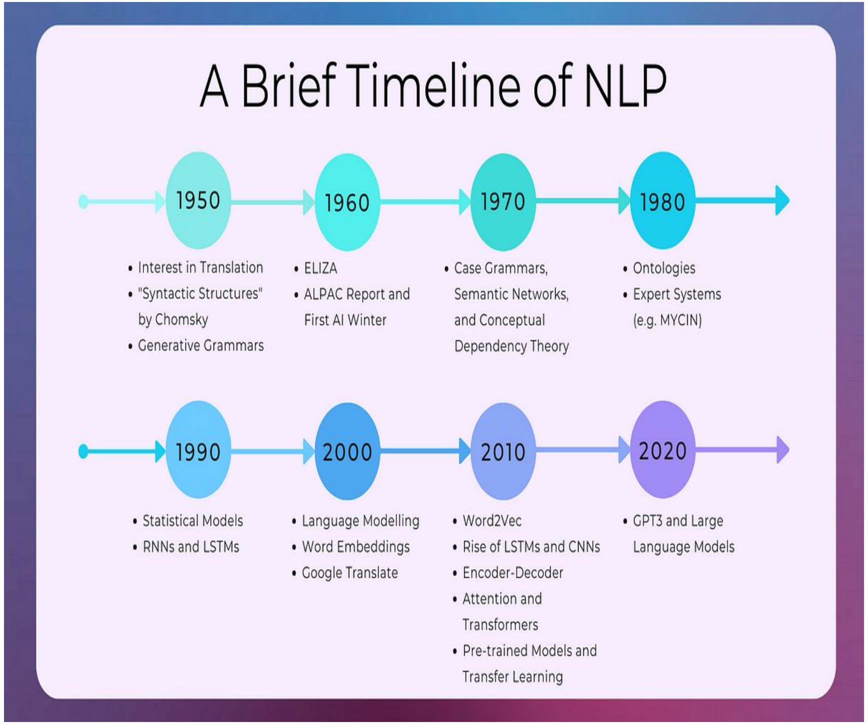
is the resultant action

The treatment

is the reward function

What is Natural Language Processing

- **Natural Language Processing (NLP)** is a field of Artificial Intelligence that enables computers to understand, interpret, and generate human language.
- Automates text understanding at scale
- Core intersection of **Linguistics**, **Computer Science**, and **AI** and bridges communication between humans and machines
- Used in in chatbots, machine translation, sentiment analysis, information retrieval
- Empowers technologies like Google Search, Siri/Alexa, Grammarly, ChatGPT



1950s–1980s	**Rule-Based Systems** (e.g., ELIZA)
1990s–2000s	**Statistical Methods** (e.g.. HMMs, n-grams)
2010s–2020s	**Deep Learning & Word Embeddings**
Today:	**Large Language Models** (BERT, GPT, etc.)

Text Preprocessing in NLP

STEP	PURPOSE	EXAMPLE
Lowercasing	Normalize text by removing caseltiv	“The Dog” → “the dog
Punctuation Removal	Eliminate non-informative symhols	Hello, world/’ → Hello world
Stop Word Removal	Remove common filler words	“This is a pen” → pen
Lemmatization/ Stemming	Reduce words to base/root form	“running”/’ran” → run

Preprocessing reduces noise and standardizes text for effective NLP modeling.

Text Representation in NLP

Converting text into numbers for machine learning models:

Bag of Words (BoW): Frequency-based; ignores context
e.g., "NLP is fun” → {"NLP":1,"'is":1, "fun':1"}

TF-IDF
- Highlights important, rare words
- Reduces weight for common words

Word Embeddings (Word2Vec, GlbVe)
- Dense vectors capture semantic similarity
- “king” – “man” + “woman” ≈ “queen”

Contextual Embeddings (BERT, GPT)
- Word meaning depends on surrounding text
- Powerful for deep language understanding

TF-IDF

A word’s importance score in a document, among N documents

When to use it? Everywhere you use “word count”,
you can likely use TF-IDF.

TF: term frequency
= #appearance in document
(high, if terms appear many times)

IDF: inverse document frequency
= log(N / #document containing the term)
(penalize “common” words appearing in almost any documents)

Final score = TF * IDF
(higher score ➡ word is more “characteristic” of document)

Tokenization

- In Natural Language Processing (NLP), a ***token*** is an individual unit of text, such as a word, punctuation mark, or subword segment used for model processing.
- The primary purpose of ***embeddings*** in Natural Language Processing (NLP) is to convert text into numerical vectors that capture semantic meaning

- Process of splitting a text object into smaller units (tokens)
- Smaller Units : words, numbers, symbols, ngrams, characters
- White space tokenizer / Unigram tokenizer

  Sentence : "I went to New-York to play football"
  Tokens : "I", "went", "to", "New-York", "to", "play", "football"
- Regular expression tokenizer

  Sentence : "Football,Cricket;Golf Tennis"

  re.split(r'[;,\s]', line)

  Tokens : "Football", "Cricket", "Golf", "Tennis"

Analytics Vidhya
Learn everything about analytics

Types of Tokenization

Word-level

- Example: 'NLP is fun" → ['NLP', "is, "fun"]

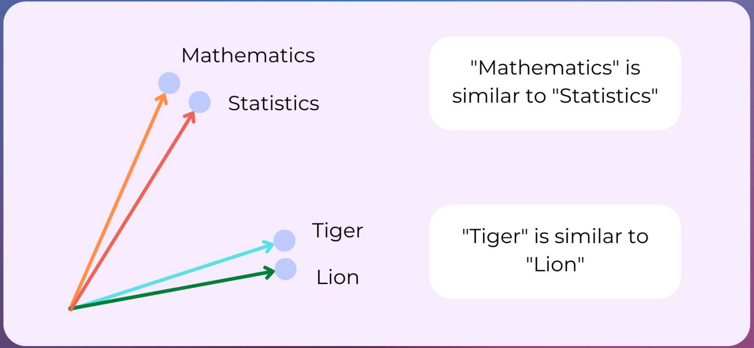
Subword-level (e.g., Byte Pair Encoding – BPE)

- Example: 'unhappiness' → ['un', 'happi,', 'ness']
- Useful for handling rare or compound words.

Sentence-level

- Example: 'NLP is fun. It powers chatbots." → ['NLP is fun.', 'It powers chatbots.']

Word Embeddings



Popular NLP Tasks

- **Text Classification**
  Sort text into categories
  e.g., spam vs not spam, sentiment analysis (*positive vs negative)*
- **Named Entity Recognition (NER)**
  Identify and classify entities like names, dates, locations in text
  Example: “Apple was founded by Steve Jobs in 1976.” → ORG: Apple, PERSON: Steve Jobs, DATE: 1976
- **Machine Translation**
  Automatically translate text from one language to another.
  Example: English ↔ French, using models like Google Translate or mBART

- **Summarization**
  Generate concise versions of longer texts.
  Extractive: Selects important sentences
  Abstractive: Generates novel, condensed versions
  e.g., news article → headline
- **Question Answering**
  Extract or generate answers from a given context.
  Example: SQuAD-style: "Who founded Apple?" → “Steve Jobs”

Language Models

- **N-gram Models:** Predict next word based on N-1 previous
- **Neural Language Models:** Use RNNs, LSTMs
- **Transformer Models:** Parallelized, self-attention-based

Examples: BERT (Encoder), GPT (Decoder), T5 (Encoder-Decoder)

NB: BERT - Bidirectional Encoder Representations from Transformers

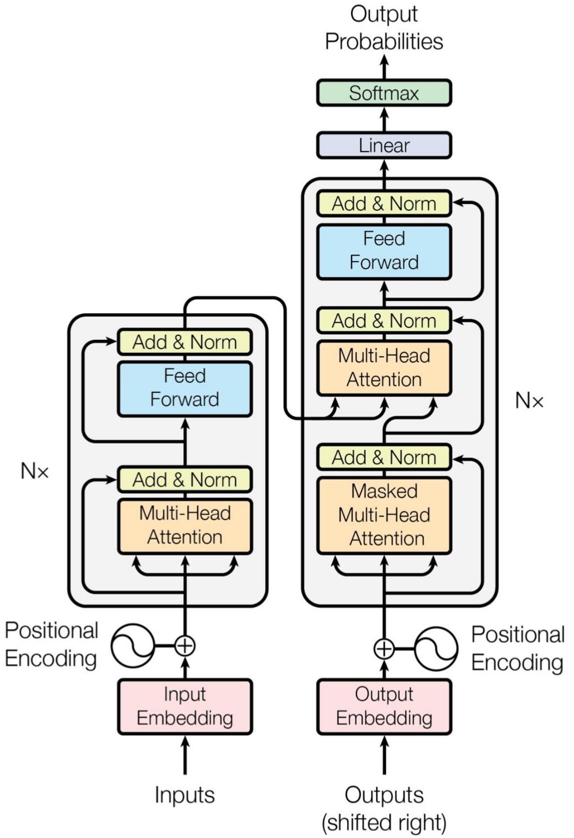
Generative Pre-trained Transformer

Text-To-Text Transfer Transformer

Model	Full Name	Architecture	Key Feature
BERT	Bidirectional Encoder Representationes from Transformers	Encoder-only	Understanding text
GPT	Generative Pre-trained Transformer	Decoder-only	Generating fluent text
T5	Text-To-Text Transfer Transformer	Encoder-Decoder	Unified format for all tasks

Generative AI & Attention Mechanism

- In **Generative AI** models like Transformers, Autoencoders and Diffusion Models, the **encoder** processes input data into a context-rich representation, while the **decoder** uses this representation to generate the output.
- **Generative AI methods** include Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Diffusion Models, Transformer-based Language Models (e.g., GPT), Autoregressive Models, etc.

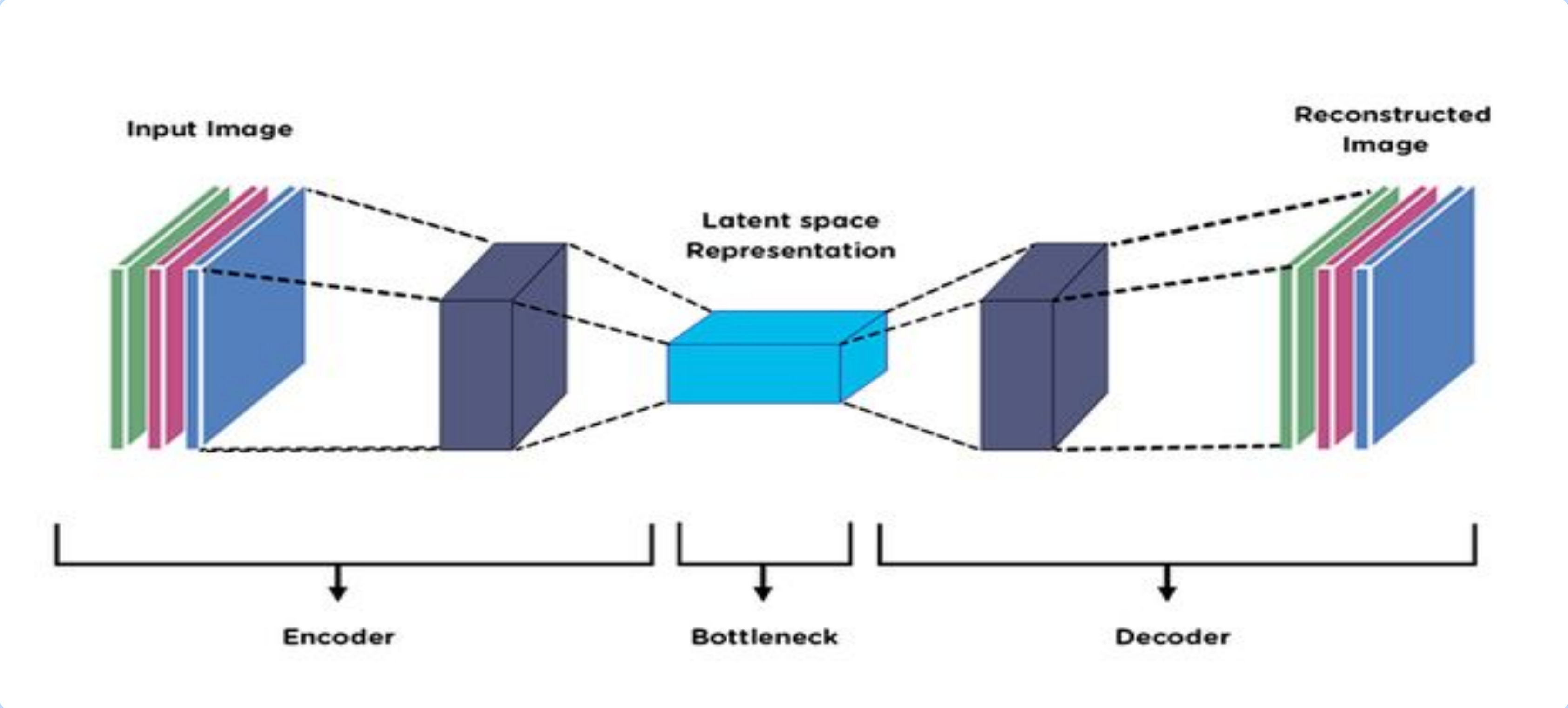


Vaswani, A., Shazeer, N., Parmar, N., et al. (2017)

Generative AI Model Architectures

Model Type	Encoder	Decoder	How It Works
cVAE (Conditional Variational Autcencender)	Yes	Yes	Maps input to a latent space, then generates conditioned output from
VAE (Variational Autoencender)	Yes	Yes	Maps input to a latent space, reconstructs original input
GAN (Generative Adversarial Network)	No	Yes	Trains a generator to produce samples that fool a discriminator
Diffusion Models	Optional	Yes	Generates data from noise by reversing a gradual noising process
Transformer-based LMs (e.g. GPT)	Yes	Yes	Generates output one token at a time in autoregressive manner

Autoencoders



Source: Premanand S (Analytics Vidhya)

Foundation Models & LLMs

- **Foundation models** are large-scale AI models trained on broad data that can be adapted to a wide range of downstream tasks
- **Foundation models** are used in:
    - In low-data situations: They can be fine-tuned with small amounts of task-specific data because they already learned general patterns from pretraining.
    - For versatility across tasks: They are trained on broad data and designed to be adapted to many tasks — even when sufficient data is available.
    - To reduce training costs: Instead of training a new model from scratch for every task, you start from a foundation model and fine-tune it — saving compute and time.
    - **Large Language Model (LLM)** is a deep learning model trained on large text corpora to understand and generate human-like language.

Evaluation Metrics

Classification

Accuracy, Precision, Recall, F1-score

Translation/Summarization

BLEU, ROUGE

Language Models

Perplexity, FID

Ranking Tasks

MAP, MRR

Evaluation Metrics in NLP

Classification

Evaluate how well model assigns correct labels

- **Accuracy** – Overall correct predictions
- **Precision** – How many predicted positives are *correctt* (e.g. "spam" predictions that were actually spam
- **Recall** – How many actual positives were found (e.g. all spam emails correctly detected)
- **F1-score** – Balance of Precision and Recall (useful with imbalanced classes)

Translation / Summarization

Measure overlap with human referernces

- **BLEU** (Bilingual Evaluation Understudy) – Measures word and phrase matches in machine translation
- **ROUGE** (Recall-Oriented Understudy for Gisting Evaluation) – Measures overlap of summaries with reference summaries (recall-based)

Language Models

Evaluate how well the model predicts word sequences

- **Perplexity** – Measures uncertainty; lower is better (e.g., how "confused" the model is when predicting the next word

Recent Trends in NLP

Stay up-to-date with where NLP is heading:

◆ **LLMs & Foundation Models**

Large-scale models trained on masive data for general language understanding and generation. Examples: GPT-4, Claude, LLaMA, LLAM, Gemini

- Versatile across tasks with little or n fine-tuning
- Serve as the backbone of modern AI systems

◆ **Prompt Engineering & In-Context Learning**

Designing inputs (prompts) that guide model behavior

- In-context learning: Model learns pattems from examples in the prompt enables zero-shot capabilities
- e,g., asking GPT to summarize, translate, or classify on—fly

◆ **Instruction Tuning**

Fine-tuning models on datasets where tasks are described via instructions

- Makes models better at following natural language commads
- Improves alignment and user intent satisfaction

◆ **Multimodal NLP**

Combining text with other data types; images, audio, video

- Enables richer understanding and generation
- Used in models like GPT-4 Vision, Gemini, and Flamingo

◆ **Responsible NLP (Bias & Fairness)**

Addressing ethical concerns in AI

- Reduce bias in training data and prdictions
- Improve fairness, safety, transparency, and accountatability

Applications in the Real World

- Healthcare (clinical notes analysis)
- Finance (document classification, fraud detection)
- Customer Service (chatbots, ticket triage)
- Education (automated grading, feedback generation
- LegalTech (contract review)

Announcement

NELIREF 4th Annual Conference comes up in August 2026 — visit our website and social media platforms for more details

https://neliref.org/

Q / A

References

1. **Online Resources** (McKinsey, AnalyticsSteps, Stanford HAI, Analytics Vidhya) with working URLs
2. **Foundational ML/AI Papers:**
   - Vaswani et al. (2017) - Attention Is All You Need
   - Devlin et al. (2019) - BERT
   - Radford et al. (2019) - GPT
   - Brown et al. (2020) - GPT-3
3. **LLM & Foundation Model Papers:**
   - Bommasani et al. (2021) - Foundation Models risks/opportunities
   - OpenAI (2023) - GPT-4 Technical Report
4. **NLP Core Methods:**
   - Mikolov et al. (2013) - Word2Vec
   - Pennington et al. (2014) - GloVe
   - Collobert et al. (2011) - NLP from Scratch
   - Goldberg (2016) - NN Architectures for NLP
   - Lample et al. (2016) - NER
5. **Generative Models:**
   - Kingma & Welling (2014) - VAEs
   - Ho et al. (2020) - Diffusion Models
   - Song et al. (2021) - Score-based Generative Modeling
6. **Advanced Topics:**
   - Wei et al. (2023) - Emergent Abilities
   - Kojima et al. (2023) - Zero-Shot Reasoning
   - Linzen et al. (2016) - LSTM Syntax Learning
   - Benchmarks: GLUE, SQuAD
7. **Textbooks:**
   - Murphy (2012) - Machine Learning: A Probabilistic Perspective
   - Goodfellow, Bengio & Courville (2016) - Deep Learning (MIT Press)
8. **Ethics & Responsible AI:**
   - Bender et al. (2021) - Stochastic Parrots (FAccT 2021)

Thank You!

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